

Pediatric Musculoskeletal Disorders

Study Guide

Name	Typical HPI	Dx	Tx
Talipes Equinovarus (aka Club Foot)	Newborn presents with hindfoot equinous and forefoot adduction	Clinical (x-rays typically not needed)	Serial casting; bracing; surgery
Metatarsus Adductus	Newborn with convexity of the lateral foot	Clinical	Reassurance; Resolves spontaneously 90%; if present > 2 years of age then serial casting; bracing; surgery
Calcaneal Apophysitis (aka Sever Disease)	9 yo F or 11 yo M, sports player, heel pain with activity	Clinical: Tenderness to palpation of posterior calcaneus, hypertonic Achilles	Rest, ice, NSAIDs, stretching, +/- heel wedges
Internal Tibial Torsion	Toddler with in-toeing	Clinical: Patient lies prone with knee flexed to 90 degrees, measure thigh-foot angle (foot will deviate toward midline)	Reassurance; Typically resolves spontaneously by 7-8 years (<1% require surgery)
**Femoral Anteversion	5 yo F with in-toeing (child may often sit in the "W" position)	Clinical: <u>Patient lies prone with knee flexed to 90 degrees, allow distal leg to move laterally</u> ; on PE entire leg appears to be internally rotated ("kissing-kneecaps")	Reassurance; Counseled to avoid sitting in "W" position; Typically resolves spontaneously by 7-8 years (<1% require surgery)
External Tibial Torsion	Child who has out-toeing that seems to worsen as they age	Clinical: have patient stand and align knees to face forward – examine foot position – feet will be laterally deviated	Reassurance; surgery if there is any dysfunction or for cosmetic concerns
Genu Valgum	3 yo with "knock-knees"	Clinical: Measure distance between the tibial medial malleoli with the knees touching	Reassurance (typically resolves by 5-8 yo); surgery if there is any pain, dysfunction or severe deformities
Genu Varum	20 month old with "bow-legs"	Clinical: Measure the distance between the medial femoral condyles with the medial malleoli touching (Need to r/o Tibia Vara)	Reassurance (typically improves by 2-3 yo); surgery if there is any pain, dysfunction or severe deformities
Tibia Vara (aka Blount Disease) (<i>Abnormal growth of medial aspect of proximal tibial epiphysis</i>)	Most commonly 2 yo obese AA F with painless "bow-legs"; could be 12 yo obese AA M with unilateral painful "bow-leg"	Weight-bearing AP & lateral x-rays of both LE (will show: Fragmentation, wedging, and beak deformities of the proximal medial tibia)	Immediate orthotics in < 3 yo Surgery > 4 yo
Popliteal Cyst (aka Baker's cyst)	10 yo with painless mass in popliteal area	Clinical (can do US)	Reassurance; surgery if progressive or limiting mobility

Osteochondritis Dissecans (Area of bone adjacent to the articular cartilage suffers a vascular insult & separates from the adjacent bone)	Child with acute knee pain and swelling	MRI	Young patient with intact articular cartilage → revascularize and heal with rest Older child with damage to cartilage → surgery
**Osgood-Schlatter Disease	11 yo F or 13 yo M basketball player with knee pain during and after activity	Clinical: tenderness and edema over tibial tubercle	Rest (stop sports while painful), NSAIDs, ice, stretching, symptoms typically last 1-2 years Complications: bony enlargement of tibial tubercle & <u>avulsion fracture</u>
Developmental Dysplasia of the Hip	Risk Factors: 1. Female (9:1) 2. Family history (20%) 3. Breech presentation 4. Firstborn child (60%) 5. Oligohydramnios	Barlow Ortolani Galeazzi If above are positive, or if F with +FHx, or any newborn born breech, then: Hip US < 4 months old Hip x-rays > 4 months old	Pavlik Harness < 6 months of age >6 months = closed reduction and hip spica cast If closed reduction fails → open reduction
Transient Synovitis	6 yo M with recent URI & acute onset of knee pain and painful limp	Need to rule out septic arthritis & osteomyelitis: +/- X-rays of joints, CBC, blood cx?	Bedrest with minimal weight bearing and NSAIDs
**Legg-Calve-Perthes Disease (avascular necrosis of the femoral head)	<u>7 yo F with atraumatic painless limp</u>	<u>AP & frog-leg x-rays of hips</u>	Pain control, restore hip motion, prevent complications; Containment (casts, orthoses, surgery)
Slipped Capital Femoral Epiphysis	Obese AA ♂ teenager with acute onset of limp and hip and/or knee pain	<u>via wheelchair</u> for AP and frog-leg x-rays of hips	Emergent surgery (internal fixation)
**Idiopathic Scoliosis (<u>most common form of scoliosis</u>)	(any age...)12 yo ♀ with R thoracic curve	Dx: clinical, Adam's forward bend test X-ray: Scoliosis series (PA & lateral films of entire spine) – Cobb angle	Tx: OMT observe if <25° brace if 25-50° >50° (or pain) = surgery
Scheuermann Kyphosis	10 yo with mid-thoracic humped back +/- pain	Forward bend; spine x-rays	Tx (depends on severity): OMT, bracing, surgery
Torticollis	Newborn with headtilt	Clinical; can do imaging if concerned it's not muscular	OMT, stretching, surgery for severe cases
**Spondylolysis (defect in pars interarticularis)/	Teen athlete with repetitive back extension & 2 weeks of LBP	Tenderness to palpation of affected vertebra, palpable step-off at LS area	Activity restriction; OMT, PT; Surgery if severe

Spondylolisthesis <i>(B/L defects with anterior slippage of the superior vertebra)</i>		AP, lateral & oblique spine x-rays; CT of spine	
Radial Head Subluxation (Nursemaid's Elbow)	Infant/Toddler with hand held in pronated position; refusal to move hand or elbow	Clinical (imaging if concern for fx)	Reduce radial head: Supinate hand while applying pressure to radial head

Barlow – infant supine, hip & knee flexed, adduct thigh with gentle downward pressure (dislocate)

- < 2-4 months

Ortolani - infant supine, hip & knee flexed, abduct thigh with gentle upward pressure (reduce)

- < 2-4 months

Galeazzi – infant supine, hip & knee flexed & feet flat on table, examine knee heights for symmetry

- >3 months

Hip pain refers to the KNEE (children with hip problems will often complain of knee pain)